Cyber Research Workshop: Student Handout

Goal: turn official cybersecurity PDFs into a dataset, use it to write a cited explainer with AI, and publish it as a website you can link on your resume.

You will make: a dataset (CSV) and a live website at `https://yourusername.github.io/cyber-research/`.

The plan

Step	What you do	Tool
1	Make a GitHub account and follow the organizer	GitHub
2	Pick a topic and open its two PDFs	Browser
3	Extract the text from a PDF (prompt 1)	Google Colab
4	Turn the text into a dataset (prompt 2)	Google Colab
5	Write a cited explainer (prompt 3)	NotebookLM
6	Turn the explainer into a website post (prompt 4)	Your AI
7	Publish it	GitHub Pages

Go at your own pace. If you get stuck on something for more than 5 minutes, ask.

Step 2: Pick a topic

Pick **one** topic. Open both PDFs. Highlight a sentence in each one: **if you can't select the text, it's a scan and won't work**, so tell an organizer.

Topic	Source A	Source B
Ransomware	CISA #StopRansomware Guide (click "Read the full guide")	NIST IR 8374: Ransomware Risk Management
Phishing	CISA Phishing Guidance: Stopping the Attack Cycle at Phase One	NIST Small Business phishing guide
AI security	OWASP Top 10 for LLM Applications 2025 (click "Download")	NIST AI 100-2 E2025: Adversarial Machine Learning
Zero trust	NIST SP 800-207: Zero Trust Architecture	CISA Zero Trust Maturity Model v2.0
Passwords and MFA	NIST SP 800-63B-4: Authentication	CISA: Implementing Phishing-Resistant MFA
Incident response	NIST SP 800-61 Rev. 3	CISA Incident and Vulnerability Response Playbooks

Topic	Source A	Source B
Software supply chain	NIST SP 800-218: Secure Software Development Framework	NTIA: Minimum Elements for an SBOM
Memory safety	The Case for Memory Safe Roadmaps	NSA: Software Memory Safety

Long PDF? Use only a range of about 20 pages (for example, pages 5 to 25). Longer takes more time to process.

Licenses (you'll need these for the dataset):

- CISA, NIST, NSA, and NTIA documents: write `US government work (verify)`
- OWASP: write `CC BY-SA 4.0`, and credit OWASP in your writeup. ShareAlike means anything you build from it should carry the same license.
- The Memory Safe Roadmaps document was co-written with international agencies, so check the copyright notice inside the PDF.

Steps 3 and 4: Google Colab

1. Go to **colab.research.google.com** and click **New notebook**.
2. Upload your PDF: click the **folder icon** on the left, then the **upload** button.
3. Open your AI of choice in another tab.

Prompt 1: extract the text

Replace the parts in [brackets], paste it into your AI, then paste the code it gives you into a Colab cell and press **Shift + Enter**.

```
I'm in Google Colab and uploaded [file.pdf] about [topic]. Write beginner-friendly
code that installs pypdf, extracts the text from pages [START] to [END] (keeping the
page number), removes repeated headers, footers, and blank lines, and previews page
1. Add short comments.
```

Prompt 2: make the dataset

```
Split that text into chunks of about 400 words with 50 words of overlap, keeping
whole sentences. Make a pandas DataFrame with these columns: id, source_name,
source_url, title, page, chunk_index, text, license, retrieved_at, topic. Use these
values: source_name = "[document name]", source_url = "[link to the PDF]", title = "
[document title]", license = "[license]", retrieved_at = "[today's date]", topic = "
[your topic]". Make the id unique, like topic-page-chunk number. Save
knowledge_base.csv, and also save knowledge_base.md with one section per chunk
(showing the source and page above each chunk's text). Show me the first 5 rows, then
download both files.
```

Check your work: look at the first 5 rows. Is the text readable? Are the page numbers right? Is anything missing? If something looks off, paste it back to your AI and say what's wrong.

Do this before closing the tab: download both files. Colab erases everything when the session ends.

Do it for your second PDF too if there's time. You can combine both into one dataset.

What the columns mean

Column	What it holds
`id`	A unique name for this chunk
`source_name`	The document it came from
`source_url`	Where to find the PDF
`title`	The document's title
`page`	The page the chunk starts on
`chunk_index`	Which chunk of the document it is (1, 2, 3...)
`text`	The text itself, about 400 words
`license`	The reuse terms
`retrieved_at`	The date you got it
`topic`	Your topic

These are the standard fields for a RAG knowledge base: a model finds the most relevant chunks and uses them to answer a question, and the source and page let you check where each answer came from.

Step 5: NotebookLM

1. Go to **notebooklm.google.com** and create a new notebook.
2. Add sources: upload `knowledge_base.md` and your PDF(s).
3. **Optional:** add 1 or 2 more reputable sources (government agencies, standards bodies, or OWASP) so you can compare perspectives. Cite them in your writeup.
4. Use this prompt in the chat:

```
Write an 800-word beginner explainer on [topic] using only the sources. Cite the
source for every claim. End with a list of things I should double-check.
```

Verify before you publish. AI can sound confident and still be wrong, which is a real risk in security.

- Click 3 of the citations and confirm the source actually says it.
- Work through the "double-check" list.
- Rewrite anything that sounds off, in your own words.

Save your finished explainer as text.

Step 6: Make it a website post

Paste this into your AI, with your explainer at the bottom:

```
Turn the writeup below into ONE Jekyll post. Output only the file contents.

Start with this front matter, exactly:
---
layout: default
title: "YOUR TITLE"
date: YYYY-MM-DD
tags: [cybersecurity]
---

Use today's date with no time. Right after the front matter, start the body with a "# YOUR TITLE" heading (the theme does not show the title by itself). Then write the rest in markdown with ## headings, and end with a "## Sources" section that links every source I used. Put any title that contains a colon inside the quotes.

Also tell me what to name the file (YYYY-MM-DD-short-title.md).

WRITEUP:
[paste your explainer here]
```

Step 7: Publish

1. Open the template: **https://github.com/n0di0/cyberramz-cyber-research**. Click **Use this template → Create a new repository**.
2. Name it exactly **cyber-research** and set it to **Public**.
3. Open `_config.yml` (pencil icon). Change the title, description, and author. You can pick a theme: keep one of the four `theme:` lines without a `#` in front (hacker, midnight, cayman, or slate). Click **Commit changes**.
4. Open the `_posts` folder, then **Add file → Create new file**. Name it like `2026-10-07-my-topic.md` and paste your post from step 6. Commit.
5. Delete the sample post (`_posts/2026-10-07-sample-post.md`): open it, click the trash icon, and commit.
6. **Settings → Pages → Deploy from a branch → `main` / `(root)` → Save.**
7. Wait 1 to 2 minutes and refresh. Your link is at the top of the Pages page.

Also upload your dataset: in your repo, **Add file → Upload files**, drag in `knowledge_base.csv`, and commit. (Later we'll put everyone's datasets together.)

If something goes wrong

The troubleshooting loop

1. Read the **last line** of the error. It usually says what's wrong.
2. Paste the **whole error and your code** into your AI and say what you expected.
3. Change **one thing at a time**, then rerun.
4. Still stuck: **Runtime → Restart session**, then run the cells again from the top.

Colab

- `FileNotFoundError`: the PDF isn't uploaded, or the file name is spelled differently. Check the folder icon.
- `ModuleNotFoundError`: run `!pip install pypdf` in a cell first.
- Empty or messy text: the PDF may be a scan. Pick another page range or another PDF.

GitHub Pages

- A 404, or "There isn't a GitHub Pages site here": wait 2 more minutes, and check that the repo is **Public** and named exactly `cyber-research`.
- The site looks plain with no styling: the repo name doesn't match. Open `_config.yml` and make sure the `baseurl` line matches your repo name, with a slash in front.
- My post doesn't show up: it must be inside `_posts`, named `YYYY-MM-DD-title.md`, dated today or earlier, and start with `---` on the first line.
- Red X on the **Actions** tab: click the failed run to read the error. It's usually a typo in the front matter.

Finished checklist

☐ `knowledge_base.csv` and `knowledge_base.md` downloaded
☐ Explainer with citations, and 3 of them checked
☐ Post is live at `https://yourusername.github.io/cyber-research/`
☐ Sources are listed at the bottom of your post
☐ Dataset uploaded to your repo